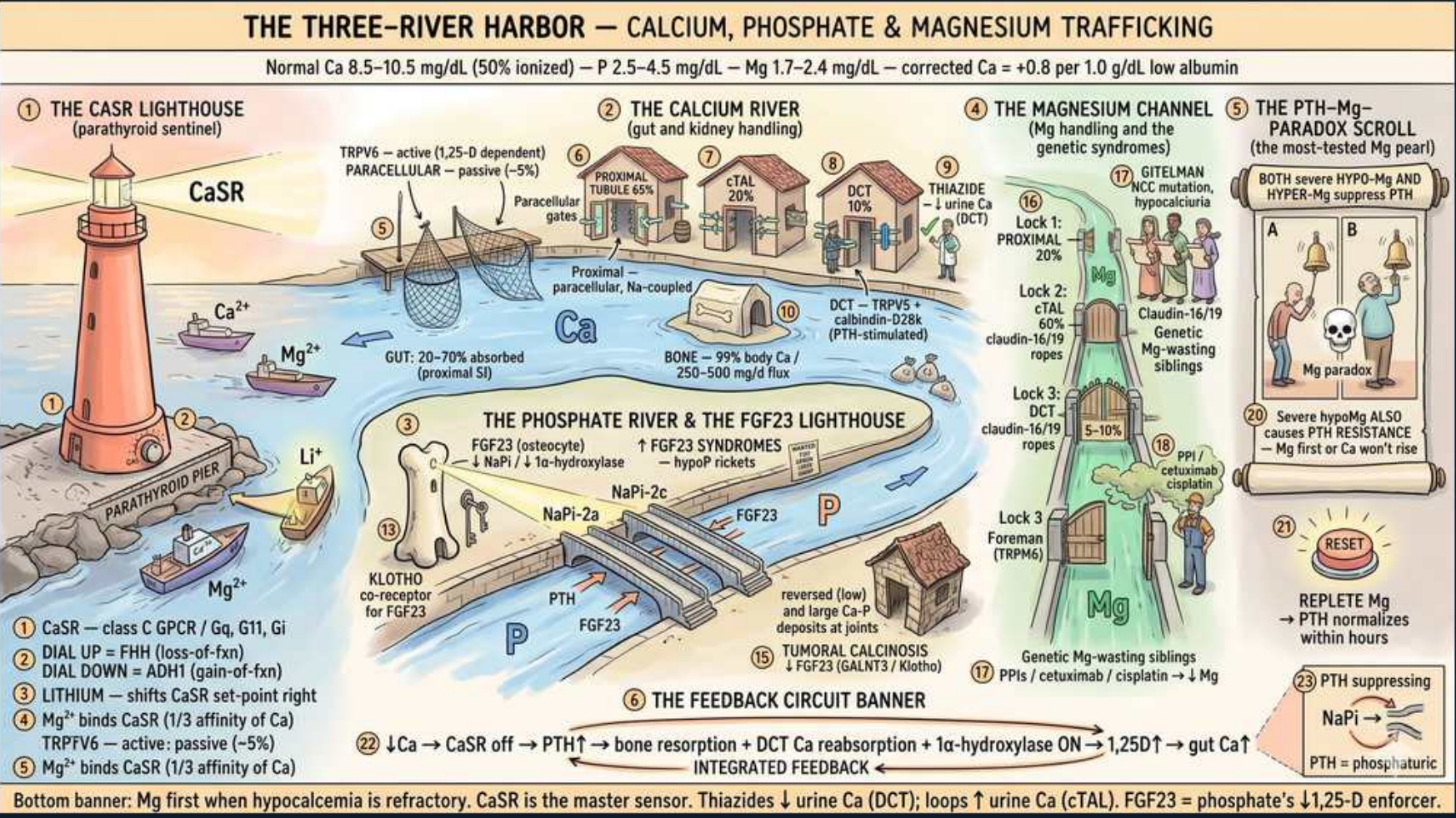


Parathyroid/Bone — Three-River Harbor: Ca, Phosphate & Magnesium



1. Normal Ca / ionized / albumin

WHAT YOU SEE

Subtitle: normal Ca 8.5-10.5 mg/dL (50% ionized); P 2.5-4.5; Mg 1.7-2.4; corrected Ca +0.8 per 1.0 g/dL low albumin

WHAT IT ENCODES

Reference ranges: total Ca 8.5-10.5 mg/dL (~50% is the biologically active IONIZED fraction, ~40% albumin-bound, ~10% complexed); phosphate 2.5-4.5 mg/dL; magnesium 1.7-2.4 mg/dL. Correct total Ca for hypoalbuminemia: ADD 0.8 mg/dL to measured Ca for every 1.0 g/dL the albumin is below 4.0. When in doubt, measure IONIZED calcium directly.

BOARD TRAP

WRONG: diagnosing hypocalcemia from a low TOTAL calcium in a hypoalbuminemic patient (cirrhosis, nephrotic, malnutrition). Low albumin lowers the bound fraction and thus total Ca, but the IONIZED (active) Ca is normal — the patient is not truly hypocalcemic. Correct for albumin or measure ionized Ca before treating.

2. CaSR lighthouse

WHAT YOU SEE

CaSR Lighthouse #1: class C GPCR (Gq/G11/Gi); the parathyroid sentinel sensing Ca

WHAT IT ENCODES

The calcium-sensing receptor (CaSR) is a class C GPCR (signals via Gq/G11 and Gi) on parathyroid chief cells and the renal tubule. It is the master Ca sentinel: HIGH serum Ca activates CaSR -> SUPPRESSES PTH secretion and increases renal Ca excretion; LOW Ca releases the brake -> PTH rises. It also senses Mg.

BOARD TRAP

WRONG: mixing up the direction of CaSR mutations. LOSS-of-function -> the gland can't 'see' high Ca -> doesn't suppress PTH -> Familial Hypocalciuric Hypercalcemia (FHH, high Ca, low urine Ca, inappropriately normal PTH) or neonatal severe hyperPTH. GAIN-of-function -> gland over-suppresses -> Autosomal Dominant Hypocalcemia (ADH, low Ca with low/normal PTH and hypercalciuria).

3. Cinacalcet dials down PTH

WHAT YOU SEE

#2: dial down -> ADH (gain-of-fxn) / cinacalcet lowers PTH set-point

WHAT IT ENCODES

Calcimimetics (CINACALCET, etelcalcetide) are allosteric CaSR activators — they 'dial down' the PTH set-point so the gland releases less PTH at any given Ca. Uses: secondary hyperPTH in dialysis CKD, parathyroid CARCINOMA, and inoperable primary hyperPTH. Pharmacologically they mimic a CaSR gain-of-function (like ADH).

BOARD TRAP

WRONG: giving cinacalcet to a hypocalcemic patient or expecting it to RAISE calcium. It LOWERS both PTH and serum Ca (CaSR activation), so it can cause/worsen hypocalcemia — it is for high-PTH states, not for treating low Ca, and not where Ca is already low.

4. Lithium raises set-point

WHAT YOU SEE

#3: LITHIUM — shifts CaSR set-point up

WHAT IT ENCODES

Lithium shifts the parathyroid CaSR set-point UPWARD — the gland now requires a higher Ca to suppress, so it keeps secreting PTH at a normal Ca -> mild HYPERcalcemia with an inappropriately normal/high PTH. Lithium also reduces urinary Ca excretion. Effect can persist and resemble primary hyperPTH.

BOARD TRAP

WRONG: diagnosing primary hyperparathyroidism in a bipolar patient on lithium without considering the drug. Lithium-induced hypercalcemia mimics primary hyperPTH (high Ca, non-suppressed PTH, low urine Ca like FHH) — review the med list; the picture may improve off lithium.

5. Calcium river handling

WHAT YOU SEE

Calcium River: gut 20-70% absorbed (proximal SI); bone 99% body Ca; paracellular vs active (1,25-D-dependent)

WHAT IT ENCODES

Calcium handling: GUT absorbs ~20-70% (proximal small intestine), via passive paracellular flux AND active transcellular transport that is 1,25-D-DEPENDENT (TRPV6/calbindin). BONE holds 99% of body Ca. KIDNEY reabsorbs ~65% in the proximal tubule (paracellular, follows Na/water) and fine-tunes in the DISTAL convoluted tubule via TRPV5, which is stimulated by PTH.

BOARD TRAP

WRONG: assuming diuretics affect Ca the same way. THIAZIDES act at the DCT to INCREASE Ca reabsorption -> RAISE serum Ca / lower urine Ca (used for recurrent Ca stones and to reduce hypercalciuria). LOOP diuretics block the TAL Na-K-2Cl transporter -> LOSE Ca in urine -> LOWER serum Ca (used in hypercalcemia after volume repletion).

6. Thiazide / loop on Ca

WHAT YOU SEE

DCT thiazide marker: thiazide up Ca reabsorption (DCT); loops down Ca

WHAT IT ENCODES

Mechanistic split: THIAZIDES (DCT) -> decrease URINARY Ca -> serum Ca rises; clinically used to prevent calcium-oxalate stones and reduce renal Ca wasting (e.g., in hypoparathyroidism). LOOP diuretics (TAL) -> increase urinary Ca -> serum Ca falls; useful adjunct in hypercalcemia (only after rehydration) or in volume overload.

BOARD TRAP

WRONG: giving a LOOP diuretic as a first move in hypercalcemia, or a THIAZIDE to a patient prone to hypercalcemia. Use a LOOP (not a thiazide) to lower Ca, and only AFTER restoring volume with saline. Conversely a thiazide can UNMASK or worsen hypercalcemia (e.g., in occult hyperPTH).

7. Phosphate / NaPi-2c

WHAT YOU SEE

Phosphate River #11: NaPi-2c reabsorption; FGF23 (osteocyte) and PTH both reduce it

WHAT IT ENCODES

Phosphate is reabsorbed in the proximal tubule by sodium-phosphate cotransporters (NaPi-2a/NaPi-2c). Two hormones promote PHOSPHATURIA by downregulating these transporters: PTH and FGF23. So serum phosphate is governed largely by renal handling, and anything raising PTH or FGF23 lowers serum phosphate.

BOARD TRAP

WRONG: thinking PTH raises phosphate the way it raises calcium. PTH raises Ca but LOWERS phosphate (it wastes phosphate in urine via NaPi downregulation). BOTH PTH and FGF23 are phosphaturic. That's why primary hyperPTH shows high Ca with LOW phosphate.

8. FGF23 / Klotho

WHAT YOU SEE

Phosphate/FGF23 Lighthouse: FGF23 needs Klotho co-receptor; FGF23 syndromes -> phosphaturic rickets

WHAT IT ENCODES

FGF23 is secreted by OSTEOCYTES and requires the co-receptor KLOTHO to signal. It does two things: increases renal phosphate EXCRETION (downregulates NaPi) and SUPPRESSES 1-alpha-hydroxylase -> lowers 1,25-D (and thus gut phosphate absorption). Net effect: lowers serum phosphate.

BOARD TRAP

WRONG: forgetting FGF23 in unexplained hypophosphatemic bone disease. FGF23 EXCESS states cause low phosphate + low 1,25-D rickets/osteomalacia: X-linked hypophosphatemia (XLH, PHEX mutation) and TUMOR-INDUCED OSTEOMALACIA (paraneoplastic FGF23 from a small mesenchymal tumor). These don't respond to plain vitamin D — need calcitriol + phosphate (or burosumab, an anti-FGF23 antibody).

9. Magnesium channel (TRPM6)

WHAT YOU SEE

Magnesium Channel: TRPM6 in DCT, Gitelman/genetic Mg-wasting siblings; loop/Lock 1-3 claudins

WHAT IT ENCODES

Magnesium is reabsorbed mostly PARACELLULARLY in the thick ascending limb (through claudin-16/19 tight junctions, driven by the lumen-positive voltage) and actively in the DCT via TRPM6. Defects or drugs here cause renal Mg wasting -> hypomagnesemia.

BOARD TRAP

WRONG: overlooking drug/genetic causes of refractory hypomagnesemia. Mg wasters: LOOP diuretics, PPIs (chronic), aminoglycosides, cisplatin, amphotericin, calcineurin inhibitors, and GITELMAN syndrome (DCT, also low K, low Ca-uria) / Bartter (TAL). Because low Mg drives refractory hypocalcemia/hypokalemia, missing it stalls correction of both.

10. PTH-Mg paradox

WHAT YOU SEE

PTH-Mg-Paradox Scroll: BOTH severe hypo-Mg AND hyper-Mg suppress PTH

WHAT IT ENCODES

The magnesium-PTH paradox: NORMAL Mg is required for PTH secretion, but BOTH severe HYPOmagnesemia AND HYPERmagnesemia SUPPRESS PTH release. Severe hypomagnesemia additionally causes end-organ PTH RESISTANCE — a double hit producing functional hypoparathyroidism and refractory hypocalcemia.

BOARD TRAP

WRONG: escalating calcium repletion in refractory hypocalcemia without checking magnesium. If Mg is low, PTH can neither be secreted nor act, so Ca will not rise — REPLETE MAGNESIUM FIRST. Refractory hypocalcemia = think (and fix) magnesium before adding more calcium.

11. Feedback circuit

WHAT YOU SEE

Feedback Circuit Banner: low Ca -> CaSR off -> PTH up -> bone resorption + DCT Ca reabsorption + 1-alpha-hydroxylase ON -> 1,25-D -> gut Ca up

WHAT IT ENCODES

Integrated calcium-restoring loop: LOW Ca -> CaSR 'off' the brake -> PTH RISES -> (1) bone osteoclastic resorption releases Ca, (2) DCT increases Ca reabsorption (and wastes phosphate via NaPi), and (3) renal 1-alpha-hydroxylase turns ON -> more 1,25-D -> gut absorbs more Ca. All three arms raise serum Ca back to normal; high Ca then re-activates CaSR and shuts PTH off.

BOARD TRAP

WRONG: forgetting that PTH's net effect on phosphate is opposite its effect on calcium. While the loop RAISES Ca via bone, kidney, and gut, the same PTH simultaneously LOWERS serum phosphate (phosphaturia) — so the classic primary-hyperPTH chemistry is high Ca + low phosphate + high PTH.

12. PTH suppressing / NaPi

WHAT YOU SEE

Bottom banner: PTH suppressing; NaPi -> down PO4; Mg first with hypocalcemia is refractory; thiazides down urine Ca; loops up urine Ca; FGF23 = phosphate / 1,25-D enforcer

WHAT IT ENCODES

Summary banner of the harbor: PTH wastes phosphate via NaPi downregulation; CaSR is the master Ca sensor (calcimimetics down, lithium up); FGF23 (+Klotho) is the phosphate/1,25-D enforcer; thiazides lower urine Ca (raise serum Ca), loops raise urine Ca (lower serum Ca); and magnesium must be corrected first in refractory hypocalcemia.

BOARD TRAP

WRONG: treating each ion in isolation. Ca, phosphate, and Mg are interdependent — fixing hypocalcemia may require correcting Mg first; lowering Ca means a loop (not thiazide) AFTER saline; and unexplained hypophosphatemia should trigger an FGF23 workup. Always read the three rivers together.